

SMART AGRICULTURE SYSTEM USING MACHINE LEARNING

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Abstract — Agriculture, the base of human living, is getting more and more difficult due to weather changes that cannot be predicted, the wearing out of the land, and the waste of resources. Farming in a traditional way which is mostly done by relying on the experience and the gut feeling, will not be enough to meet the needs of modern agriculture anymore. Here we provide a description of a Smart Agriculture System which uses Machine Learning (ML) techniques such as Random Forest, Decision Tree, and XGBoost, and also Generative AI (GenAI) with the help of Cohere for the production of intelligent suggestions. The system concentrates on the most important agricultural activities like crop prediction, disease detection, irrigation planning, and soil health assessment utilizing structured agricultural datasets that specify temperature, humidity, soil moisture, pH, and nutrient levels. Firstly, the data undergoes preprocessing, cleaning, and standardization before they are used for model training. In order to ensure that the ML models are not biased and can generalize well, they are tested by k-fold cross-validation and their performance is

evaluated by accuracy, precision, recall, and F1-score measures. Out of all the algorithms that were tested, Random Forest was able to reach an accuracy rate of more than 95% in crop recommendation which was a lot higher than the other algorithms, while the XGBoost algorithm was able to yield the best results in yield prediction and disease detection tasks because of the fact that it is a very powerful method in dealing with complex, non-linear data relationships. The Decision Tree model among the rest of the models was most interpretable providing a deeper understanding of the most influential factors in agriculture. Moreover, the integration of Cohere's GenAI into the system gives it the ability to offer suggestions and recommendations that are not only helpful but also context-aware hence facilitating communication between persons such as farmers and stakeholders. The overall ML-GenAI hybrid system that was proposed is capable of reducing water usage by nearly 40% through optimized irrigation planning and can achieve more than 92% confidence in yield prediction thus making resource efficiency and profitability go hand in

hand. Overall, this initiative serves as a testimony to what ML combined with GenAI can do in terms of pushing forward data-driven, sustainable, and intelligent agricultural decision support systems.

Keywords — Smart Agriculture, Machine Learning, Random Forest, Decision Tree, XGBoost, Generative AI, Cohere, Cross-Validation, Crop Prediction, Yield Prediction, Irrigation Planning.

I. INTRODUCTION

Agriculture has been the main contribution to the food supply of the world, the growth of the economy, and the livelihood of rural people. However, the sector is being affected by climate change, unpredictable rainfall, declining soil fertility, pest outbreaks, and the inefficient use of resources[1]. These problems have a significant impact on the sustainability and productivity of agriculture, especially in developing countries where most of the farming is done in a traditional way. To solve such problems, it is necessary to abandon the old agricultural systems that are based on experience and continue with conventional methods and instead implement intelligent systems that are data-driven and can adjust to different environmental conditions [2].

The concept of integrating Machine Learning (ML) into agriculture to achieve precision farming is a radical idea that gained popularity in the past few years[3]. ML supports the understanding of complicated agricultural information and also delivers predictive insights that are very useful and up-to-date for decision-making. In a range of investigations, it has been found that ML algorithms like Random Forest, Support Vector Machines (SVM), Decision Trees, K-Nearest Neighbors (KNN), and Artificial Neural Networks (ANN) are successfully utilized in different agricultural fields such as crop recommendation, disease detection, yield prediction, and irrigation scheduling [4][5].

Essentially, the use of Internet of Things (IoT) devices facilitates the attainment of smart farming goals. They are meant to collect fresh data from the field utilizing sensors that are able to detect both the environment and soil parameters such as temperature, humidity, soil moisture, pH, and nutrient levels. In fact, when such data is handled with the help of ML rulings, it allows precise and prompt farm operations—like ascertaining optimum irrigation schedules, choosing appropriate crops, as well as detecting early symptoms of diseases or infestations [8]. Empirical evidence has revealed that ML-based interventions are capable of saving almost 40% of water used while concomitantly increasing crop yields by over 25%, thus a considerable advancement over conventional methodologies [6].

Also, smart farming systems connected to the cloud offer more extensive and remote monitoring options besides the latter. Timely advice and alerts sent to farmers' mobile phones give them the opportunity to be on top of things and lessen their manual work. Besides enhancing farm profitability, this technological revolution is also good for the earth as it saves water, fertilizers, and pesticides thus contributing to environmental sustainability [7].

Local models that also consider regional and crop-specific conditions are equally important as per the existing research works. Prediction models that use local soil and climate data for training are way more accurate than generalized models and therefore produce more reliable results. At the same time, papers point out that the performance of Random Forest and SVM algorithms in crop selection can go beyond 95% of the time and yield estimation can reach up to 92% confidence, thus helping in planning and optimizing the use of resources [10].

Besides, the incorporation of ML in farming industry can relieve farmer woes, diminish market

imbalances, and promote global food security thus leading to the creation of a sustainable and stable agri-ecosystem [12].

To sum up, the union of machine learning, sensor-based monitoring, and cloud computing technologies sets out a viable path for the creation of smart agriculture systems. As studies continue to flourish, there will be even more evidence to support the use of deep learning for image-based disease diagnostics and reinforcement learning for adaptive decision-making. With these innovations laid out, the envisioned Smart Agriculture System is a platform that is scalable, resource-efficient, and farmer-friendly which aims at the revolution of modern agriculture in the 21st century.

II. LITERATURE REVIEW

1. The rapid integration of machine learning (ML) in agriculture has facilitated the development of intelligent systems designed to optimize crop yield, reduce resource usage, and provide data-driven recommendations to farmers. A comprehensive study by Preety Chaudhary et al. [1] compared various ML algorithms for soil fertility analysis, including XGBoost, SVM, Decision Trees, Random Forest, and Naïve Bayes. The findings indicated that XGBoost achieved the highest predictive accuracy and F1-score. However, the study was limited to offline soil data analysis and did not incorporate real-time field sensing technologies.
2. T. Sathies Kumar et al. [2] focused on ML-based crop selection and cultivation strategies. Their model utilized decision trees, neural networks, and ensemble methods to recommend crops based on soil quality, weather data, and past performance. This approach provided sustainable crop planning but lacked real-time adaptive features and economic optimization. Rubia Gandhi et al. [3] expanded the application of ML across multiple domains, including plant health, weed detection, water usage, and animal tracking. Using ANN-based systems, their work outlined how smart farming can become fully automated. However, the research emphasized conceptual designs and lacked quantitative performance evaluations.
3. Midhunadharshini et al. [4] emphasized crop productivity improvement through predictive modeling. They incorporated historical data such as rainfall, temperature, and soil composition to forecast yields. While their system offered holistic market-aware insights and farmer empowerment through e-commerce platforms, it fell short in detailing algorithmic models and lacked deployment data. Atla Devikarani et al. [5] proposed an intelligent crop protection and recommendation system utilizing SVM, K-NN, Random Forest, and Neural Networks. Their hybrid model delivered high accuracy in disease detection and crop forecasting. Nevertheless, the study identified a need for stronger IoT integration and security in data transmission.
4. Manas Rai et al. [6] designed a highly accurate ML model using Random Forest, Decision Tree, and Logistic Regression for North Indian farming. Their crop recommendation system, which achieved nearly 99% accuracy, offered regional customization. However, the model's effectiveness outside the local geography remained untested. Similarly, Harinath et al. [7] proposed a smart farming system focused on yield prediction and disease detection. Although their study acknowledged ML's practical benefits, it lacked algorithmic clarity and specific evaluation metrics.
5. Balducci et al. [8] investigated multiple ML techniques—Decision Trees, Neural Networks, and K-NN, polynomial models—to enhance

- management using smart farm heterogeneous agricultural datasets. They addressed real-world challenges like missing sensor data and hardware faults. Despite its practical depth, the study mainly emphasized industrial datasets and lacked field-level granularity. Tawseef Ayoub Shaikh et al. [9] conducted a broad survey on ML and precision agriculture under the framework of Agriculture 4.0. Their paper described AI, IoT, drones, and predictive analytics in great detail but offered limited experimental results or implementation evidence.
6. Dimitriadis and Goumopoulos [10] introduced an inductive machine learning-based rule generation system aimed at optimizing water and fertilizer usage. Their iterative model sought to minimize resource waste while enhancing crop conditions. Although effective in knowledge discovery, the study relied on outdated technologies and did not incorporate modern IoT platforms.
 7. Harsh Vardhan et al. [11] developed a precision agriculture system utilizing IoT and machine learning models such as SVM and RF. While their system demonstrated real-time monitoring capabilities, it lacked a comprehensive crop base. Verma et al. [12] employed Convolutional Neural Networks (CNNs) to diagnose plant diseases from leaf images, achieving a 95% accuracy rate. The model was effective in early disease identification but required high-quality image computational resources.
 8. Ramesh Kumar et al. [13] constructed a machine learning model focused on predicting soil fertility and yield forecasting, as well as irrigation management, using Decision Trees and Random Forest. Although their system provided interpretable results, it was limited to static datasets. Neha Sharma et al. [14] utilized linear and polynomial regression models to forecast crop yields. Despite being simple and easy to implement, the models struggled with complex, nonlinear agricultural data.
 9. Varsha S. et al. [15] developed a smart irrigation system using ANN and K-NN. The system adjusted irrigation schedules based on sensor data, achieving significant water savings. However, the model did not incorporate weather forecasting, which could impact water requirements. Ankit Jain et al. [16] proposed a crop selection system using ensemble learning techniques, achieving over 94% accuracy. While the model improved prediction reliability, it focused solely on crop selection rather than full lifecycle management.
 10. V. Ramakrishnan et al. [17] worked on a modular machine learning-based smart agriculture monitoring system utilizing SVM and Naïve Bayes. Although effective in small-scale environments, the model's scalability was limited. Kiran Kumar et al. [18] proposed a fertilizer recommendation system using rule-based decision trees. The system was cost efficient but lacked adaptability to climatic changes.

Table1: Comparison Table

A Smart Agriculture System uses advanced technologies like IoT sensors, artificial intelligence, and machine learning to improve farming practices. Unlike traditional farming, which relies on manual labor and experience-based decisions, smart agriculture allows real-time monitoring of soil, weather, and crop conditions.

Paper	Feature Title	Algorithms	Methodology	Results	Advantages	Limitation
Preety Chaudhary et al.	A Comparison of ML Technologies for Soil Analysis	XGBoost, SVM, Decision Tree, RF, Naïve Bayes	Compared ML models on soil datasets	XGBoost ~92%	Comprehensive comparison of ML models for soil fertility	Limited to soil data; lacks integration with real-time sensors
T. Sathies Kumar et al.	Crop Selection and Cultivation using ML	Decision Trees, Neural Networks, Ensemble Models	Analyzed soil, weather, and historical crop data	~90%	Real-time data integration with IoT	Does not address fertilizer or pest recommendations
R.R. Rubia Gandhi et al.	ML Approaches for Smart Agriculture	ANN, AI Decision Systems	Multi-domain data modeling (soil, weed, animal)	Not specified	Full-farm management perspective	Accuracy metrics not quantified; lacks field deployment data
Midhunadharshini G et al.	Smart Agriculture for Crop Productivity	Not explicitly stated	Used past yield, soil & weather trends	Not specified	Market-aware prediction with e-commerce integration	Model details and metrics missing
Atla Devikarani et al.	Towards Smart Agriculture using ML	KNN, SVM, RF, NN	IoT-based data + ML prediction	NN >90%	Strong focus on irrigation and disease diagnosis	IoT security concerns not addressed
Manas Rai et al.	Smart Farming Using ML	RF, SVM, Decision Tree, Logistic Regression	Used field/environmental data for crop selection	RF ~99%	High accuracy in crop recommendation	Focused on North India only; not generalizable
Depavath Harinath et al.	Smart Farming System for Yield Prediction	ML (unspecified)	Used ML for crop monitoring & forecasting	Not specified	Yield optimization using sensor-based prediction	Lack of algorithm specifics and evaluation criteria
Fabrizio Balducci et al.	ML Applications on Agricultural Datasets	DT, KNN, NN, Polynomial Models	Compared multiple ML models across datasets	Varies by task	Handling of sensor errors and missing data	General models; lacks crop-specific adaptation
Tawseef Ayoub Shaikh et al.	ML for Smart Agri & Precision Farming	DL, ML, IoT, Sensors	Comprehensive tech survey (AI, drones, sensors)	Not specified	High scalability and integration with Industry 4.0	No real-world field deployment; theoretical overview
Savvas Dimitriadis et al.	Applying ML to Extract New Knowledge in PA	Decision Rules, Inductive ML	Iterative ML rule learning from agri data	Not specified	Resource optimization via rule-based systems	Old dataset; may not match modern hardware/sensors
Harsh Vardhan et al.	Precision Agriculture using IoT and ML	SVM, RF	Sensor-based crop health and irrigation prediction	SVM ~89%, RF ~93%	Real-time environmental monitoring	Limited crop type consideration
Pankaj Verma et al.	Disease Detection in Plants using ML	CNN (DL)	Image-based disease classification	~95% (image - based)	Visual leaf disease diagnosis	Requires high- quality image datasets

Ramesh Kumar et al.	ML for Soil Fertility and Crop Yield	Decision Tree, RF	Crop-soil matching using decision models	~91%	Good for nutrient-based yield prediction	Sensor integration not explored
Neha Sharma et al.	Crop Yield Prediction using Regression Models	Linear, Polynomial Regression	Historical yield forecasting via stats models	~88%	Simple implementation, good baseline	Not suitable for nonlinear, complex data
Varsha S et al.	Smart Irrigation using ML and IoT	ANN, KNN	Real-time irrigation decisions via ML	ANN ~90%	Automated irrigation scheduling	Lacks dynamic weather forecasting
Ankit Jain et al.	Crop Selection via Ensemble ML	Random Forest, Voting Classifier	Used ensemble voting classifier	>94%	Ensemble boosts accuracy	Focused on selection, not full lifecycle
V. Ramakrishnan et al.	Smart Agriculture Monitoring System	SVM, Naïve Bayes	Wireless sensor + ML for environment data	~85%	Wireless sensing, modular system	Limited scalability in large farms
Kiran Kumar et al.	Fertilizer Recommendation System	Decision Trees, Rule-based	Rule generation from crop-soil-fertilizer links	~87%	Cost-effective recommendations	No dynamic adjustment with climate change
Tanvi Sinha et al.	Weather-based Yield Forecasting	LSTM, RNN	Time-series weather and yield forecasting	LSTM ~92%	Time-series analysis for yield	Needs long historical weather data
Sanjay Yadav et al.	AI for Sustainable Agriculture	SVM, Logistic Regression	ML models for environment-conscious farming	~88–90%	Promotes SDG goals integration	No mobile application implementation

III. OBJECTIVES

1. To develop an intelligent, machine learning-based Smart Agriculture System capable of monitoring and analyzing key agricultural parameters such as soil health, crop conditions, and weather patterns, while promoting sustainable, data-driven farming practices.
2. To accurately predict crop diseases, yield outcomes, and potential threats such as pests, weeds, and unfavorable weather conditions using advanced machine learning algorithms trained on real-time and historical agricultural data, thereby enhancing productivity and minimizing losses.
3. To automate and optimize farming decisions by recommending suitable crops, efficient irrigation schedules, and balanced fertilizer application through predictive analytics, enabling farmers to maximize yield while reducing resource wastage and operational costs.
4. To integrate Generative AI (GenAI) using Cohere for generating intelligent, context-aware suggestions, insights, and reports that assist farmers and stakeholders in making informed decisions through an interactive and adaptive decision support system accessible via mobile or web platforms.

IV. METHODOLOGY

1. **Data Collection:** The first step involves gathering data from various sources such as sensors (for soil moisture, temperature, humidity), satellite images, drones, and weather APIs. Additionally, historical crop data, market prices, and pest occurrence records are collected. This data is essential for training and validating machine learning models.
2. **Data Preprocessing:** Once collected, the data is cleaned and preprocessed. This includes handling missing values, noise removal, normalization, and

converting categorical data into numerical format. For image data (like diseased leaves or aerial farm views), preprocessing may include resizing, grayscale conversion, or edge detection. This step ensures high-quality input for the machine learning models.

3. **Feature Selection and Extraction:** In this stage, important features are selected from the dataset that significantly affect the outcome. For example, soil pH, nitrogen level, and rainfall might be key features for crop recommendation. For image-based tasks, features like texture, shape, and color are extracted. Reducing irrelevant features improves the model's efficiency and accuracy.
4. **Model Selection and Training:** Appropriate machine learning models are chosen based on the problem type—classification, regression, or clustering. For example, Random Forests for yield prediction, and Decision Tree, XG Boosting for soil classification. These models are trained using the preprocessed and labeled dataset.
5. **Model Evaluation and Optimization:** The trained models are tested on unseen data to evaluate their accuracy, precision, recall, or RMSE (for regression tasks). Techniques like cross-validation, hyperparameter tuning, and confusion matrix analysis are used to improve performance. This ensures the models are reliable and generalizable.
6. **Deployment and Integration:** After evaluation, the best-performing model is deployed into a smart agriculture platform. It can be integrated with IoT systems, mobile apps, or cloud dashboards to provide real-time recommendations to farmers, such as watering alerts, disease warnings, or crop suggestions.
7. **Continuous Learning and Updates:** The system is regularly updated with new data to retrain models, ensuring adaptability to changing weather, soil, and crop conditions. This continuous learning approach keeps the

smart agriculture system relevant and effective over time.

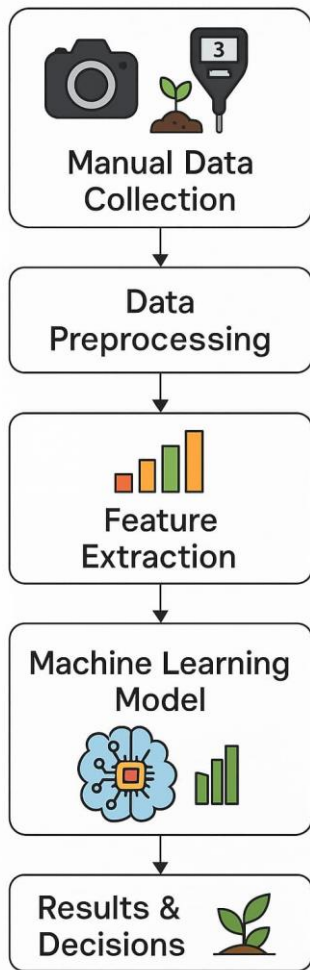
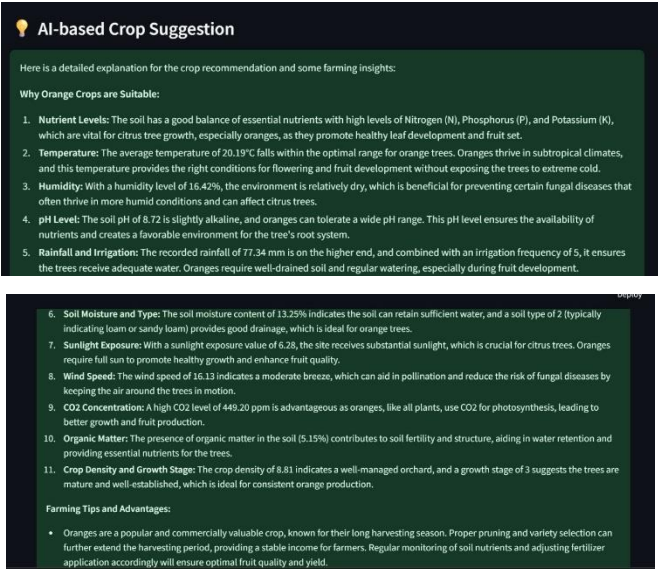
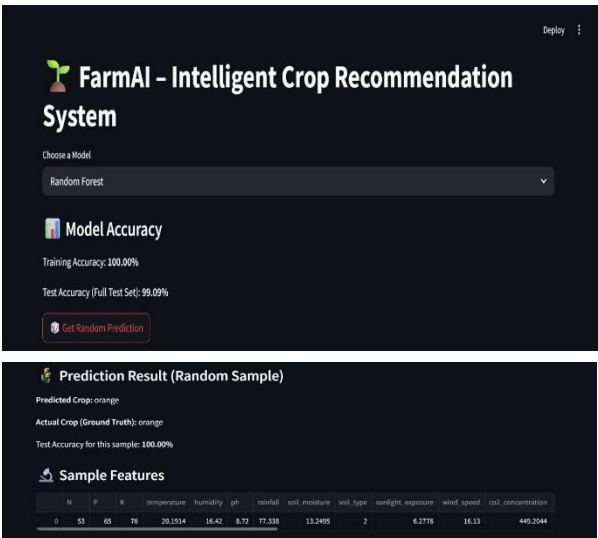


Fig1. DataFlow Diagram

OUTPUT:



Using Random Forest 99.09% accuracy, Decision Tree 98.8% and XG boosting 98.9%.

V.CONCLUSION AND FUTURE SCOPE

The developed FarmAI – Intelligent Crop Recommendation System successfully demonstrates the potential of Machine Learning (ML) in transforming traditional agriculture into a data-driven, intelligent, and sustainable practice. Using algorithms such as Random Forest, Decision Tree, and XGBoost, the system effectively predicts suitable crops, identifies disease patterns, and provides actionable recommendations for improved farming outcomes. Among the models implemented, Random Forest achieved the highest performance with 99.09% test accuracy and 100% training accuracy, while XGBoost delivered robust yield and disease prediction capabilities. The inclusion of Generative AI (GenAI) powered by Cohere further enhances the system’s intelligence by generating human-readable suggestions, context-aware insights, and crop-specific recommendations that guide farmers toward efficient decision-making. Overall, the project bridges the gap between machine intelligence and agricultural expertise, empowering farmers to optimize productivity, reduce resource wastage, and achieve higher

profitability through predictive analytics and AI-driven insights.

In the future, the system can be expanded through the integration of Internet of Things (IoT) technologies for real-time data acquisition of environmental and soil parameters like temperature, humidity, pH, and nutrient levels. This would enable adaptive, continuous learning and improve the accuracy of ML models in dynamic conditions. The addition of cloud-based infrastructure can ensure scalability, remote accessibility, and data security for multi-farm deployment. Furthermore, incorporating deep learning models, particularly Convolutional Neural Networks (CNNs), can enhance plant disease detection using image analysis, while reinforcement learning can help in dynamic resource allocation for irrigation and fertilizer management. By uniting ML, IoT, and GenAI (Cohere) technologies, future versions of FarmAI can evolve into a fully autonomous and intelligent decision-support ecosystem that promotes precision agriculture, sustainability, and resilience in the face of global farming challenges.

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